

PART A — QUESTIONS

AF24-001

Question:

Solve: $26x + 16 = 172$

- A) 5
- B) 6
- C) 7
- D) 8

AF24-002

Question:

If $24n - 17 = 151$, what is n ?

- A) 5
- B) 6
- C) 7
- D) 8

AF24-003

Question:

Simplify: $11(3x - 4) - 10x$

- A) $23x - 44$
- B) $33x - 44$
- C) $23x - 4$
- D) $43x - 44$

AF24-004

Question:

A company charges \$55 per order plus a \$24 handling fee. Which expression represents the total charge for o orders?

- A) $55o + 24$
- B) $24o + 55$
- C) $79o$
- D) $55 - 24o$

AF24-005

Question:

If $f(x) = 21x - 75$, what is $f(8)$?

- A) 83
- B) 93
- C) 168
- D) 243

AF24-006

Question:

Which ordered pair satisfies the equation $y = 19x - 8$?

- A) (1, 12)
- B) (2, 29)
- C) (3, 49)
- D) (4, 70)

AF24-007

Question:

Solve: $30(x - 8) = 210$

- A) 13
- B) 14
- C) 15
- D) 16

AF24-008

Question:

A pattern begins: 18, 41, 64, 87, ... What is the next number?

- A) 108
- B) 109
- C) 110
- D) 111

AF24-009

Question:

Solve: $x/60 = 3$

- A) 63
- B) 120
- C) 180
- D) 240

AF24-010

Question:

A line has a slope of -21 and crosses the y-axis at 18. Which equation represents the line?

- A) $y = 18x - 21$
- B) $y = -21x + 18$
- C) $y = 21x + 18$
- D) $y = -18x + 21$

AF24-011

Question:

If 7 bins contain 322 parts, how many parts are in 4 bins at the same rate?

- A) 172
- B) 180
- C) 184
- D) 192

AF24-012

Question:

Solve: $500 - 45x = 320$

- A) 3
- B) 4
- C) 5
- D) 6

AF24-013

Question:

Which expression is equivalent to $144p - 67p + 16$?

- A) $77p + 16$
- B) $211p + 16$
- C) $77p - 16$
- D) $144p - 51$

AF24-014

Question:

A warehouse starts with 2,800 packages. Each truck removes 175 packages. Which equation gives the number of packages P left after t trucks?

- A) $P = 2,800t - 175$
- B) $P = 175t + 2,800$
- C) $P = 2,800 - 175t$
- D) $P = 175t - 2,800$

AF24-015

Question:

If $y = -35x + 550$, what is y when $x = 14$?

- A) 50
- B) 60
- C) 490
- D) 1,040

AF24-016

Question:

Solve: $42x - 110 = 30x + 46$

- A) 11
- B) 12
- C) 13
- D) 14

AF24-017

Question:

Which value of x makes $26x + 12$ greater than 350?

- A) 11
- B) 12
- C) 13
- D) 14

AF24-018

Question:

A table shows a linear relationship.

x : 0, 1, 2, 3

y : 7, 31, 55, 79

What is the rule?

- A) $y = 7x + 24$
- B) $y = 24x + 7$
- C) $y = 31x - 24$
- D) $y = x + 7$

AF24-019

Question:

Solve: $26(x + 4) = 23x + 146$

- A) 12
- B) 13
- C) 14
- D) 15

AF24-020

Question:

If $a = -19$ and $b = 16$, what is $4a + 5b$?

- A) -156
- B) -4
- C) 4
- D) 156

AF24-021

Question:

Which point is located in Quadrant III?

- A) (20, -11)
- B) (-20, 11)
- C) (-20, -11)
- D) (20, 11)

AF24-022

Question:

A worker labels 330 boxes in 15 minutes. At the same rate, how many boxes can the worker label in 8 minutes?

- A) 160
- B) 168
- C) 176
- D) 184

AF24-023

Question:

Solve for x : $G = 48x + 96$

- A) $x = G - 96$
- B) $x = (G - 96)/48$
- C) $x = 48(G - 96)$
- D) $x = G/48 + 96$

AF24-024

Question:

What is the slope of a line that goes through (12, 25) and (18, 127)?

- A) 15
- B) 16
- C) 17
- D) 102

AF24-025

Question:

Simplify: $108x - 45 - 97x + 66$

- A) $11x + 21$
- B) $205x + 21$
- C) $11x - 111$
- D) $205x - 111$

AF24-026

Question:

If $g(x) = x^2 - 14x$, what is $g(18)$?

- A) 54
- B) 72
- C) 252
- D) 324

AF24-027

Question:

A sequence follows the rule “multiply by 5, then subtract 19.” If the first term is 12, what is the second term?

- A) 31
- B) 41
- C) 60
- D) 79

AF24-028

Question:

Solve: $30(x - 6) + 50 = 320$

- A) 13
- B) 14
- C) 15
- D) 16

AF24-029

Question:

A line crosses the y-axis at -27 and rises 13 units for every 1 unit to the right. Which equation represents the line?

- A) $y = -27x + 13$
- B) $y = 13x - 27$
- C) $y = -13x - 27$
- D) $y = 27x + 13$

AF24-030

Question:

If $40/60 = x/180$, what is x ?

- A) 100
- B) 110
- C) 120
- D) 130

AF24-031

Question:

Solve: $-60x + 300 = -600$

- A) -15
- B) -10
- C) 10
- D) 15

AF24-032

Question:

A stockroom starts with 9,600 connectors. Each batch uses 480 connectors. Which expression gives the number of connectors left after b batches?

- A) $9,600 + 480b$
- B) $9,600 - 480b$
- C) $480b - 9,600$
- D) $9,600 \div 480b$

AF24-033

Question:

Which equation has $x = 34$ as its solution?

- A) $2x + 22 = 90$
- B) $5x - 60 = 115$
- C) $x/17 + 8 = 11$
- D) $3x - 20 = 84$

PART B — ANSWER KEY + EXPLANATIONS

AF24-001 — Correct: B — Explanation:

Solve $26x + 16 = 172$. Subtract 16 from both sides: $26x = 156$. Divide by 26: $x = 6$. The most common mistake is subtracting 16 correctly but forgetting to divide by 26.

AF24-002 — Correct: C — Explanation:

Start with $24n - 17 = 151$. Add 17 to both sides: $24n = 168$. Divide by 24: $n = 7$. A common mistake is subtracting 17 instead of adding it.

AF24-003 — Correct: A — Explanation:

Distribute first: $11(3x - 4) = 33x - 44$. Then subtract $10x$: $33x - 10x - 44 = 23x - 44$. A common mistake is forgetting to multiply -4 by 11.

AF24-004 — Correct: A — Explanation:

The company charges \$55 for each order, so that part is $55o$. The \$24 handling fee is added one time. The total charge is $55o + 24$. A common mistake is treating the handling fee as the per-order charge.

AF24-005 — Correct: B — Explanation:

Substitute 8 for x : $f(8) = 21(8) - 75 = 168 - 75 = 93$. A common mistake is adding 75 instead of subtracting it.

AF24-006 — Correct: C — Explanation:

Test the choices in $y = 19x - 8$. For $(3, 49)$, $19(3) - 8 = 57 - 8 = 49$, which matches the y -value. A common mistake is choosing a point that is close but does not make the equation true.

AF24-007 — Correct: C — Explanation:

Solve $30(x - 8) = 210$. Divide both sides by 30: $x - 8 = 7$. Add 8: $x = 15$. A common mistake is adding 8 before undoing the multiplication by 30.

AF24-008 — Correct: C — Explanation:

The pattern increases by 23 each time: 18, 41, 64, 87. Add 23 to 87: 110. A common mistake is assuming the pattern increases by 22 because the numbers are close together.

AF24-009 — Correct: C — Explanation:

$x/60 = 3$ means x divided by 60 equals 3. Multiply both sides by 60: $x = 180$. A common mistake is adding 60 and 3 instead of multiplying.

AF24-010 — Correct: B — Explanation:

Slope-intercept form is $y = mx + b$. The slope is -21 and the y-intercept is 18, so the equation is $y = -21x + 18$. A common mistake is switching the slope and y-intercept.

AF24-011 — Correct: C — Explanation:

Find the unit rate: $322 \text{ parts} \div 7 \text{ bins} = 46 \text{ parts per bin}$. For 4 bins: $46 \times 4 = 184$. A common mistake is multiplying 322 by 4 without finding the rate first.

AF24-012 — Correct: B — Explanation:

Solve $500 - 45x = 320$. Subtract 500 from both sides: $-45x = -180$. Divide by -45: $x = 4$. A common mistake is losing the negative sign after subtracting 500.

AF24-013 — Correct: A — Explanation:

Combine like terms: $144p - 67p = 77p$. The constant 16 stays the same. The expression becomes $77p + 16$. A common mistake is combining the constant with the variable terms.

AF24-014 — Correct: C — Explanation:

The warehouse starts with 2,800 packages. Each truck removes 175 packages, so t trucks remove $175t$ packages. The number left is $P = 2,800 - 175t$. A common mistake is adding $175t$ even though packages are being removed.

AF24-015 — Correct: B — Explanation:

Substitute $x = 14$ into $y = -35x + 550$: $y = -35(14) + 550 = -490 + 550 = 60$. A common mistake is treating $-35(14)$ as positive 490.

AF24-016 — Correct: C — Explanation:

Solve $42x - 110 = 30x + 46$. Subtract $30x$: $12x - 110 = 46$. Add 110: $12x = 156$. Divide by 12: $x = 13$. A common mistake is adding 110 to only one side.

AF24-017 — Correct: D — Explanation:

Solve $26x + 12 > 350$. Subtract 12: $26x > 338$. Divide by 26: $x > 13$. The only listed value greater than 13 is 14. A common mistake is choosing 13, but $26(13) + 12 = 350$, not greater than 350.

AF24-018 — Correct: B — Explanation:

The y -values increase by 24 each time x increases by 1, so the slope is 24. When $x = 0$, $y = 7$, so the y-intercept is 7. The rule is $y = 24x + 7$. A common mistake is using 7 as the slope because it is the first y -value.

AF24-019 — Correct: C — Explanation:

Distribute: $26x + 104 = 23x + 146$. Subtract $23x$: $3x + 104 = 146$. Subtract 104: $3x = 42$. Divide by 3: $x = 14$. A common mistake is forgetting to multiply 4 by 26.

AF24-020 — Correct: C — Explanation:

Substitute $a = -19$ and $b = 16$: $4a + 5b = 4(-19) + 5(16) = -76 + 80 = 4$. A common mistake is treating $4(-19)$ as positive 76.

AF24-021 — Correct: C — Explanation:

Quadrant III has a negative x-coordinate and a negative y-coordinate. The point $(-20, -11)$ fits this description. A common mistake is choosing $(-20, 11)$, which is Quadrant II.

AF24-022 — Correct: C — Explanation:

330 boxes in 15 minutes means $330 \div 15 = 22$ boxes per minute. In 8 minutes: $22 \times 8 = 176$. A common mistake is multiplying 330 by 8 without finding the rate first.

AF24-023 — Correct: B — Explanation:

Start with $G = 48x + 96$. Subtract 96 from both sides: $G - 96 = 48x$. Divide by 48: $x = (G - 96)/48$. A common mistake is dividing G by 48 first and then subtracting 96.

AF24-024 — Correct: C — Explanation:

Slope equals change in y divided by change in x . From $(12, 25)$ to $(18, 127)$, y increases by 102 and x increases by 6. The slope is $102 \div 6 = 17$. A common mistake is using 102 as the slope without dividing by the change in x .

AF24-025 — Correct: A — Explanation:

Combine like terms: $108x - 97x = 11x$, and $-45 + 66 = 21$. The simplified expression is $11x + 21$. A common mistake is writing $11x - 21$ because the positive 66 is missed.

AF24-026 — Correct: B — Explanation:

Substitute 18 for x : $g(18) = 18^2 - 14(18) = 324 - 252 = 72$. A common mistake is calculating 18^2 as 36 instead of 324.

AF24-027 — Correct: B — Explanation:

Start with 12. Multiply by 5: 60. Then subtract 19: 41. A common mistake is subtracting 19 first and then multiplying.

AF24-028 — Correct: C — Explanation:

Solve $30(x - 6) + 50 = 320$. Distribute: $30x - 180 + 50 = 320$. Combine: $30x - 130 = 320$. Add 130: $30x = 450$. Divide by 30: $x = 15$. A common mistake is forgetting to combine -180 and $+50$.

AF24-029 — Correct: B — Explanation:

The line rises 13 units for every 1 unit to the right, so the slope is 13. It crosses the y -axis at -27 , so the equation is $y = 13x - 27$. A common mistake is switching the slope and intercept.

AF24-030 — Correct: C — Explanation:

Use the proportion $40/60 = x/180$. Since $60 \times 3 = 180$, multiply 40 by 3: $x = 120$. A common mistake is adding 120 to the numerator because 60 becomes 180.

AF24-031 — Correct: D — Explanation:

Solve $-60x + 300 = -600$. Subtract 300 from both sides: $-60x = -900$. Divide by -60: $x = 15$. A common mistake is giving -15 because both sides contain negative numbers.

AF24-032 — Correct: B — Explanation:

The stockroom starts with 9,600 connectors. Each batch uses 480 connectors, so b batches use $480b$ connectors. The number left is $9,600 - 480b$. A common mistake is adding $480b$ even though connectors are being used.

AF24-033 — Correct: A — Explanation:

Test $x = 34$. In choice A, $2(34) + 22 = 68 + 22 = 90$, so the equation is true. A common mistake is choosing a nearby equation without checking both sides exactly.